

Lab 5: Visualization

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1 Task 1: Parallel Coordinates Visualization

We implemented the parallel coordinates visualization with performance comparable to the given demo. The color scheme is similar to the demo. We also added interactive features, which allow users to select ranges on each axis to filter the displayed data points.

The update is lazy and is only triggered when the user releases the mouse button after selecting a range. However, since the entire visualization is built on a CPU-based backend, the operation is not very smooth.

The interactive features are listed below:

- **Range Selection:** Click and drag on an axis to select a range of values. Data points outside this range will be filtered out.
- **Reset Selection:** Double-click on an axis to reset the selection and show all data points for that axis.
- **Drag Axes:** Click and drag an axis to reorder it within the parallel coordinates plot.

The interaction is slightly laggy due to the CPU-based rendering, but it still provides a basic level of interactivity for exploring the dataset.

The results are shown in Figure 1.

2 Task 2: Flow Visualization

We implemented a basic flow visualization using Line Integral Convolution (LIC).

The functions `GetVector` and `GetNoise` are used to interpolate the vector field and noise texture using bilinear interpolation.

The function `compute_integral_curve` is the core function that computes the integral curve starting from a given point in both forward and backward directions. The step size is $1.0f$. The total curve length is parameter L , with forward and backward lengths being $L/2$ each. We clamped the positions to the boundary of the vector field to avoid invalid access.

We use a box filter to average the noise values along the integral curve. The results are shown in Figure 2.

Note that we must normalize the vector field before computing the integral curve to ensure consistent step sizes. This is very important for correct visualization.

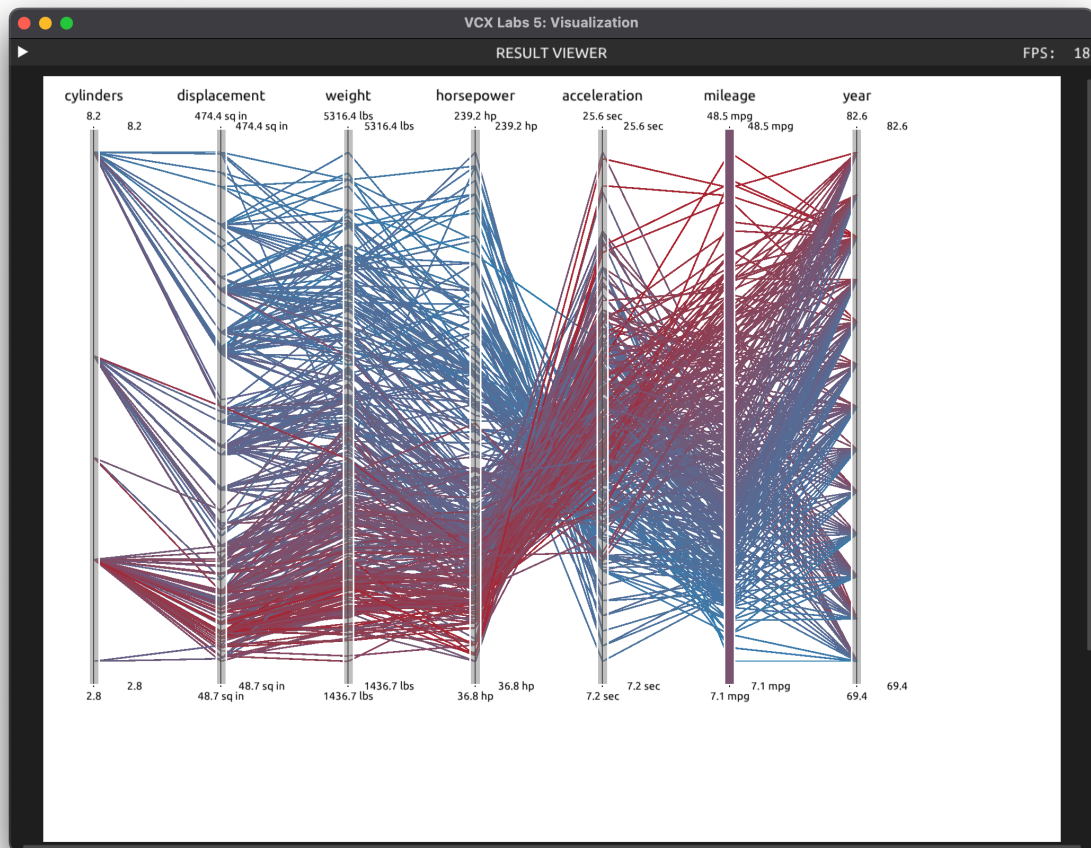


Figure 1: Parallel Coordinates Visualization

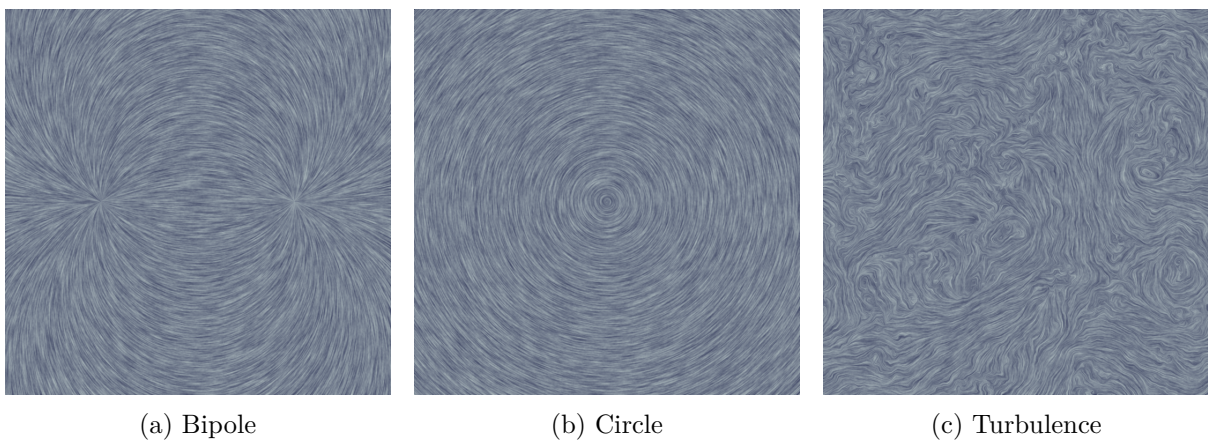


Figure 2: Flow visualization using LIC with step size 25